



Fig. 8: Elements of Sea Sparrow missile system

Missile launchers are situated at two places in the carrier; one is on the left side front of the carrier and the other is on the right or left side aft of the carrier. Each launcher carries eight missiles in ready-to-fire mode. SAM, on the side of the incoming cruise missile, is fired to intercept the threat. Also, the carrier is manoeuvred so that more SAM launchers face the side from where maximum number of cruise missiles are incoming. This creates a crossfire area where two SAM systems can target the threats.

Evolved Sea Sparrow missile. The evolved Sea Sparrow SAM has a semi-active radar-guidance system with midcourse datalink update scheme, like SM-2 SAM. It has an interception range of more than 50km.

Through AN/SPN-48 radar, high-flying targets are identified and tracked. The engagement passes on to Mark-23 TAS. For low-flying cruise missiles, Mark-23 TAS directly comes into play.

Mark-23 TAS. Mark-23 is a 2D pulse-Doppler acquisition radar. It operates in 1GHz–2GHz (L band) and has been specifically designed to operate in high-clutter environments. The radar can detect a target with a radar cross-section (RCS) as mini-

mum as 1m at ranges around 40km. Generally, cruise missiles have low RCS. The radar automatically starts tracking the fast-approaching cruise missiles.

This radar can also be used for aircraft traffic control in a secondary role. In this role, the range is around 185km.

This radar is gyro-stabilised and can tackle 30° pitches and rolls. It provides 360° coverage and up to 75° coverage in elevation. It rotates at 15rpm during normal azimuthal scanning. But under threat scenarios, it can rotate at 30rpm to track the target intensely. It can simultaneously track 54 targets. The desired target to be intercepted is chosen in the FCS. Then, illumination radars come into play being initiated by the FCS.

Mark-95 illuminator. This radar operates in X band and produces 2kW average power. Generally, X band radiators are capable of producing a thin beam, which is essential for accuracy and resolution. This radar houses transmitter and receiver antennae separately. Cylindrical structures are radomes. Concave and convex radomes enclose receiver and transmitter antennae respectively. Between these two is an electrooptical imager for electrooptical tracking,

as a backup arrangement. There are four Mk-91 illuminators in a carrier.

In a saturation cruise missile strike against the carrier, there are many cruise missiles flying towards the carrier. There is a serious possibility that one or two cruise missiles escape interception from Sea Sparrow air defence envelope. The last resort is the ship's self-defence system (SSDS).

The SSDS

SSDS is the second line of the carrier's air defence. For a carrier there is a possibility of at least one incoming cruise missile in all eight directions during a saturation cruise missile strike. In this scenario, even a second lost in decision making implies destruction. So what claims importance here is prioritising threats and engaging these sequentially and, most importantly, quickly. For this, a decision-making automatic C² system is required. SSDS is a system to neutralise these low-flying cruise missiles.

This is not a new system; it is simply a fibre-optic local area network (LAN) around an SSDS C² system, connected with self-defence weapon systems like rolling airframe missile (RAM) and close-in weapon system (CIWS) called Phalanx, and sensors like long-range air-search radar AN/SPS-49, surface-search radar AN/SPS-67 and a centralised IFF system, and an electronic warfare suite.

The irony of SSDS is that, this integration in no way augments the individual system's efficiency from its standalone capability. But integration reduces the reaction time during complicated cruise-missile-attack profiles. Data from all sensors are clubbed to form a single tactical picture in the computer. Tracks are analysed, threats are identified from the tracks. From here, threats are prioritised and finally engaged through a suitable weapon system.

For example, if two cruise missiles are approaching the carrier from different directions, the one closer to the carrier is given priority and engaged.